
TRAVAUX DIRIG  S N   3 : Linear Separators

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EXERCICE 1. We work in the framework of binary classification : let X be a random descriptor taking values in $\mathbb{R}$, equipped with its Borel sigma-algebra, and let Y be a random label taking the value 0 or 1. Let $\mathcal{G} := \{g : \mathbb{R} \rightarrow \{0, 1\}\}$ be the set of classifiers suited to this setting. The classification error is defined as the map $L : g \in \mathcal{G} \mapsto \mathbb{P}(Y \neq g(X)) \in [0, 1]$, and we write $L^* := \inf_{g \in \mathcal{G}} L(g)$.

In this exercise, we consider the family $\mathcal{G}_0$ of linear classifiers on $\mathbb{R}$ of the form :

$$g_{(x_0, y_0)} : x \in \mathbb{R} \mapsto \begin{cases} y_0 & \text{if } x \leq x_0, \\ 1 - y_0 & \text{otherwise,} \end{cases}$$

where $(x_0, y_0) \in \mathbb{R} \times \{0, 1\}$. The classification error of such a $g_{(x_0, y_0)}$ is denoted more simply by $L(x_0, y_0)$, and we set $L_0 := \inf_{(x_0, y_0)} L(x_0, y_0)$.

1) Express the classification error of an arbitrary element of $\mathcal{G}_0$ in terms of the conditional distributions of X given Y . Use the notation $F_y(x) := \mathbb{P}\{X \leq x \mid Y = y\}$ for $(x, y) \in \mathbb{R} \times \{0, 1\}$ and $p := \mathbb{P}(Y = 1)$.

2) By considering the points $(x_0, y_0) = (-\infty, 0)$ and $(x_0, y_0) = (-\infty, 1)$, show that $L_0 \leq \frac{1}{2}$.

3) Show that $L_0 = \frac{1}{2} - \sup_x \left| p F_1(x) - (1 - p) F_0(x) - p + \frac{1}{2} \right|$. Simplify the expression when $p = \frac{1}{2}$.

Hint. For every $(a, b) \in \mathbb{R}^2$, one can write $\min(a, b) = \frac{a + b - |a - b|}{2}$.

4) Show that $L_0 = \frac{1}{2}$ if and only if $L^* = \frac{1}{2}$.

5) Prove the Chebyshev–Cantelli inequality : for every real-valued random variable Z and every $t \geq 0$,

$$\mathbb{P}(Z - \mathbb{E}(Z) \geq t) \leq \frac{\mathbb{V}(Z)}{\mathbb{V}(Z) + t^2}.$$

6) Let m_y and σ_y^2 denote respectively the expectation and variance of the conditional distribution of X given $Y = y$, with $y \in \{0, 1\}$. Show that :

$$L_0 \leq \left(1 + \frac{(m_0 - m_1)^2}{(\sigma_0 + \sigma_1)^2} \right)^{-1}.$$

Hint. Use the inequality proved in the previous question.

7) Discuss the performance of the empirical minimizer taken over the class $\mathcal{G}_0$ and the limitations of linear classifiers.